Robotic Locomotion & Kinematics & Wall Following

Armando Sousa

Jorge Ribeiro

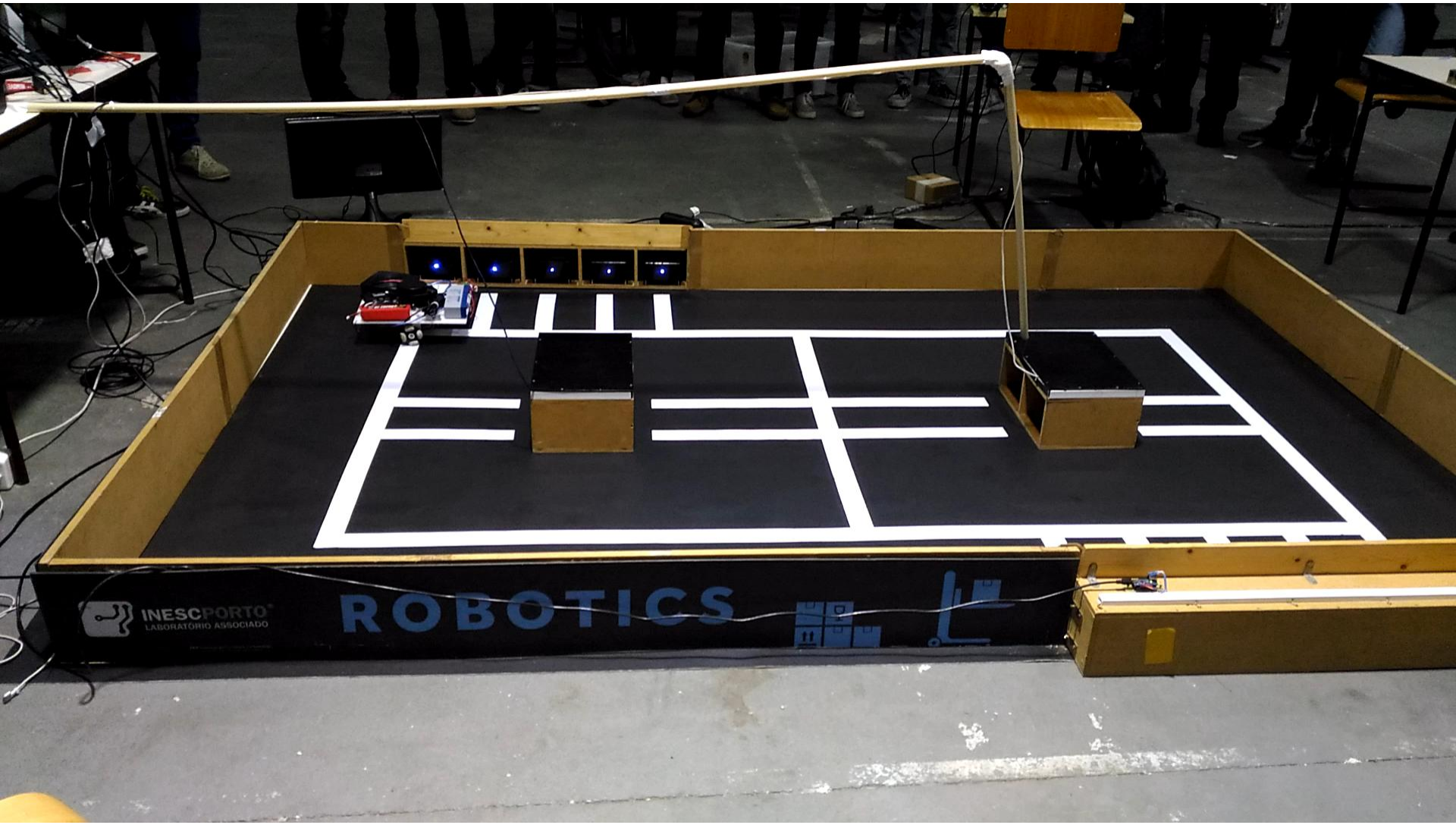
Locomotion

(Mobile Robot)

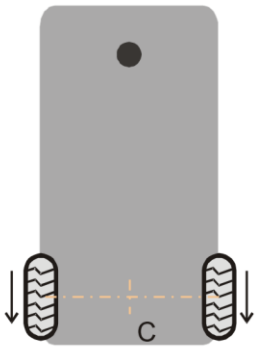
(motivational video)

https://youtu.be/C_tiur59Q2E

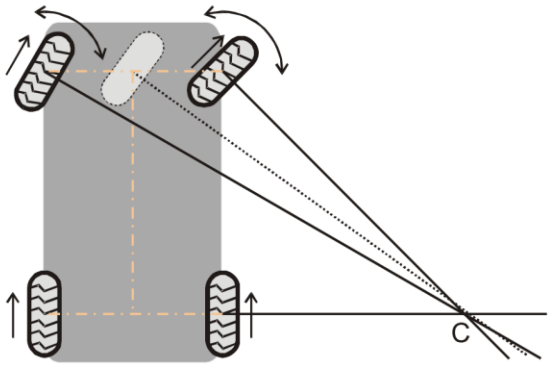
Robot at Factory



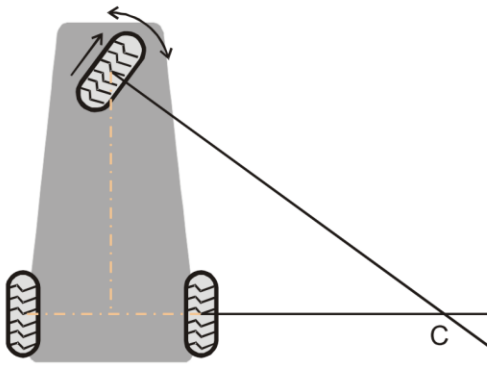
Locomotion (i) - Wheels



Differential drive
(wheelchair)

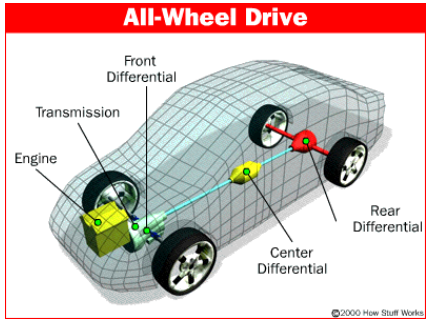


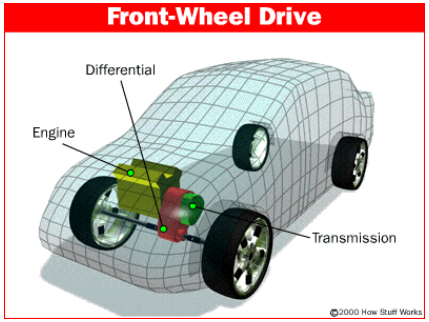
Ackerman
(normal car drive)

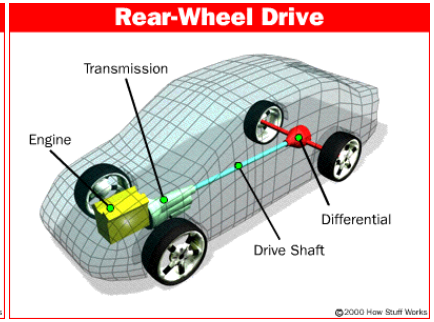


Tricycle

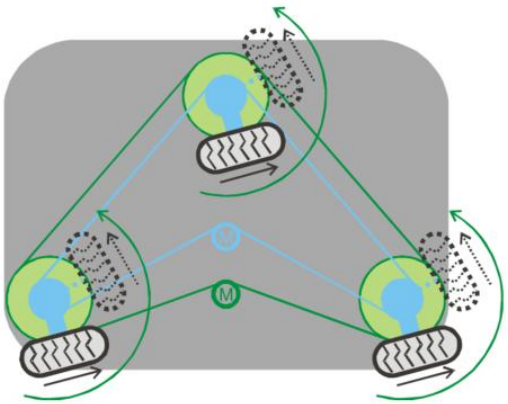
Ackerman
(normal car drive)





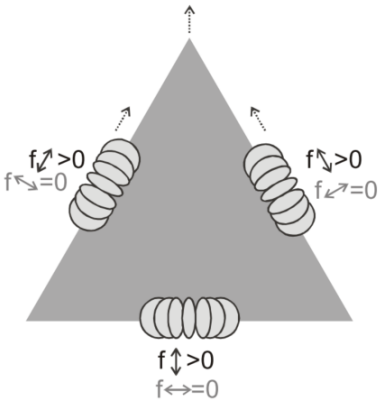


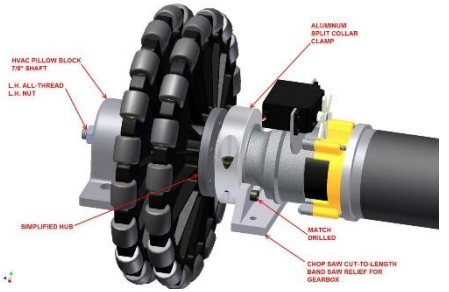
Locomotion (ii) – Omni / Omniwheels

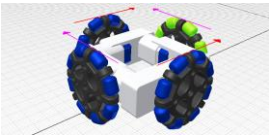


Synchronous
(Quasi omnidrive)

Locomotion (iii) – Omni / Omniwheels

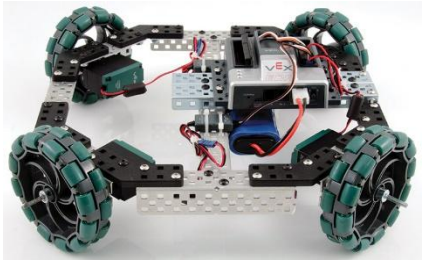


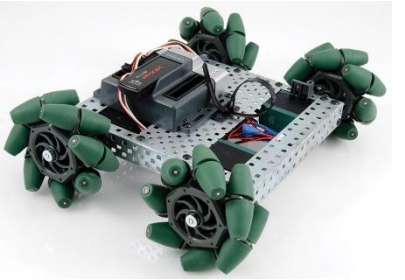




Omni Wheels Holonomic Drive

Mecanum Wheel





https://www.servomagazine.com/magazine/article/a-look-at-holonomic-locomotion
https://www.chiefdelphi.com/t/help-with-omni-wheels/83764/21

https://www.vicenzathunders.com/products

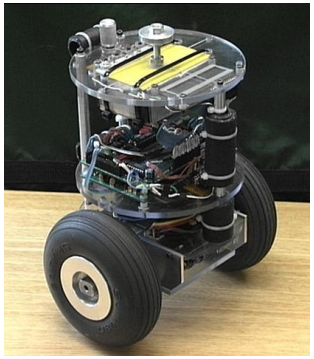
https://youtu.be/noqBUEgyQ8A

https://youtu.be/EG2pka4Bczg

Mecannum (Omni) Wheels

- http://youtu.be/o-j9TReI1aQ
- http://youtu.be/Yng7JB6swuI
- http://youtu.be/TXTo16KKm8Q
- Omni Lifter Airtrax Cobra - http://youtu.be/IlmKcohyXG0
- Lego Omni Wheel - http://youtu.be/7fevklHUaIk
- https://youtu.be/fzZEWjjmYNM

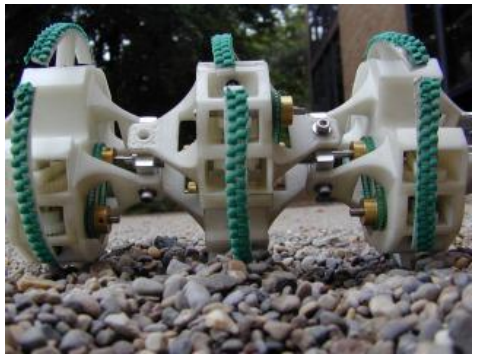
Locomotion (iv)



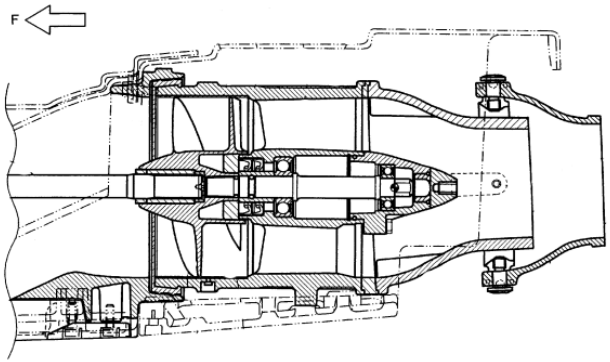
Nbot / segway

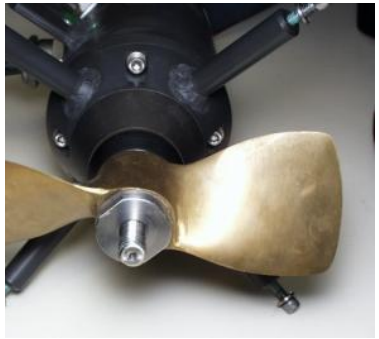
Track Drive
(military tank like)

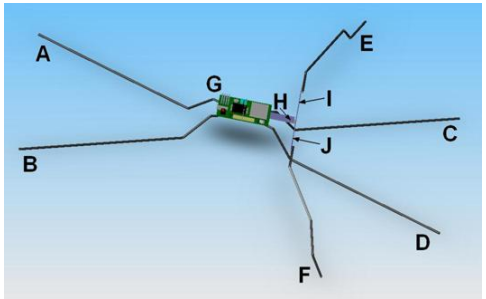
Soryu



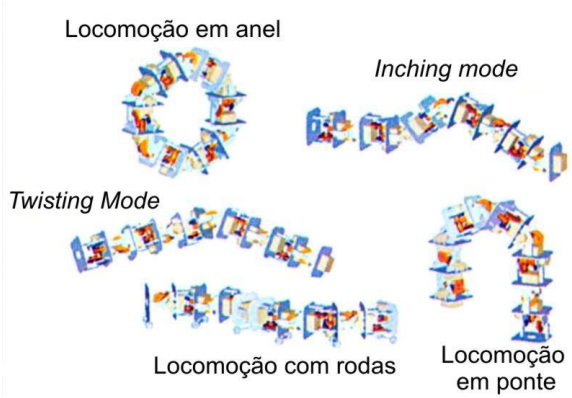
Locomotion (v) – water related

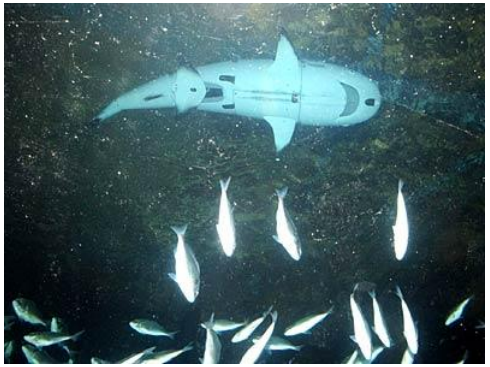




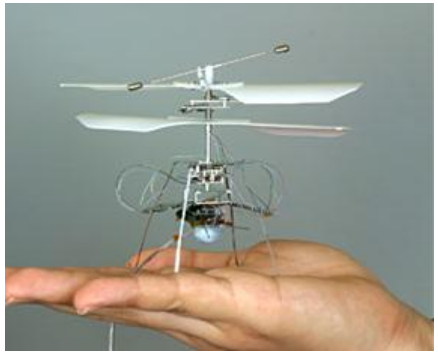


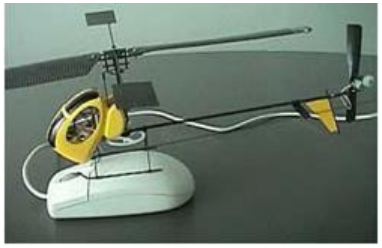
Locomotion (vi)





Locomotion (vii) – Heli, Jet, Mars, other…





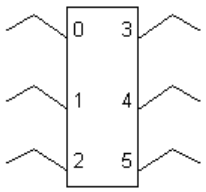
Other motion principles

- Jet / Propellers (air/water)
  - Dynamic (ruder, plane wings - *pressure difference*)
  - Helicopter
  - Wings / directive fins / rudders (air/water)
- Rocket / Solar Wind / Ion Thrusters
- Wind...
- ...
- Keep a sharp eye –

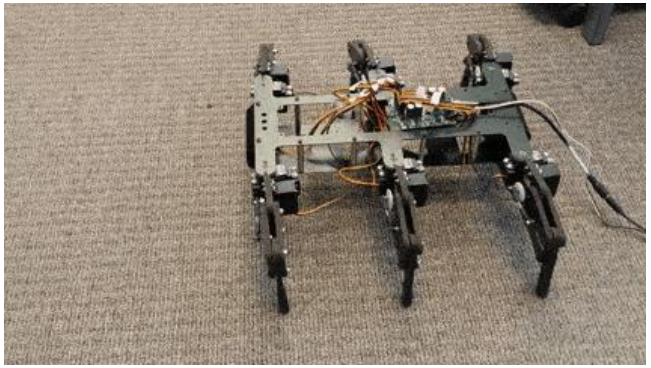
  many, MANY other options !!!

Legged Locomotion Example: 6 legs

Forward Sequence:
- Lift legs 0, 2 and 4.
- Pull back legs 1, 3 and 5 to push robot forwards.
- Lower legs 0, 2 and 4.
- Lift legs 1, 3 and 5.
- Move legs 1, 3 and 5 forward to reset them.
- Pull back legs 0, 2 and 4 to push robot forwards.
- Lower legs 1, 3 and 5.
- Lift legs 0, 2 and 4.
- Move legs 0, 2 and 4 forward to reset them.
- Lower legs 0, 2 and 4.



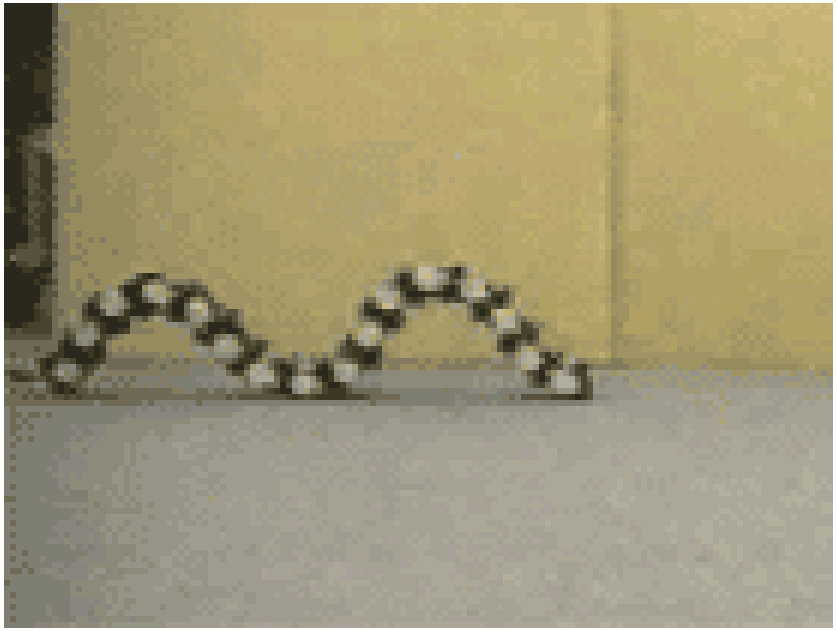
Turn Right Sequence:
- Lift legs 1, 3 and 5.
- Push back legs 0 and 2 to turn robot.
- Lower legs 1, 3 and 5.
- Lift legs 0 and 2.
- Move legs 0 and 2 forward to reset them.
- Lower legs 0 and 2.



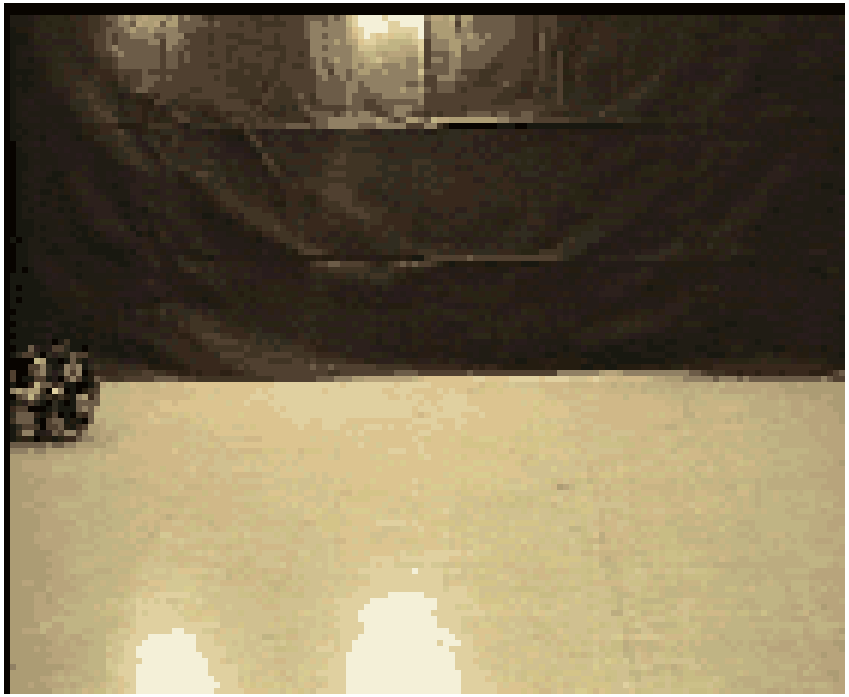
http://www.mcdonald.org.uk/andrew/archive/robot/comp/movement.htm
http://www.youtube.com/watch?v=ZhB4nFN_mRI
http://youtu.be/-Soq9qpK5Ac

https://www.phidgets.com/?view=articles&article=HexWalkerFirstSteps

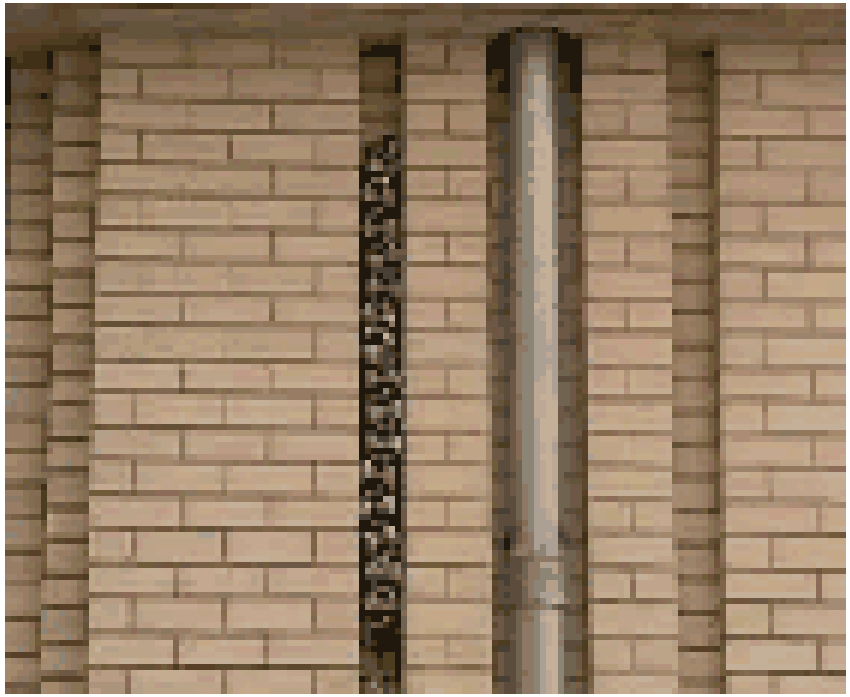
Locomotion animations



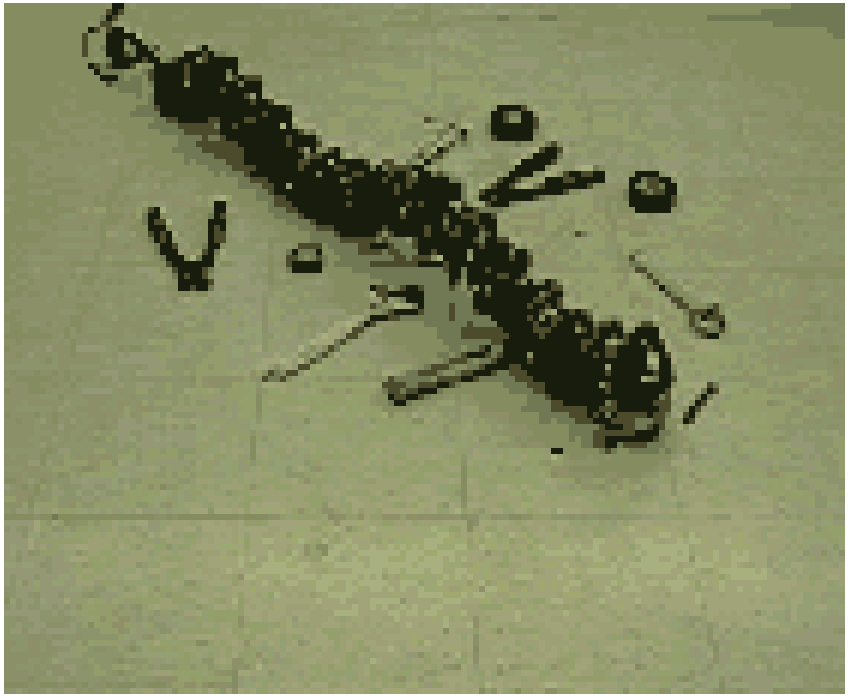
Locomotion animations (ii)



Locomotion animations (iii)



Locomotion animations (iv)



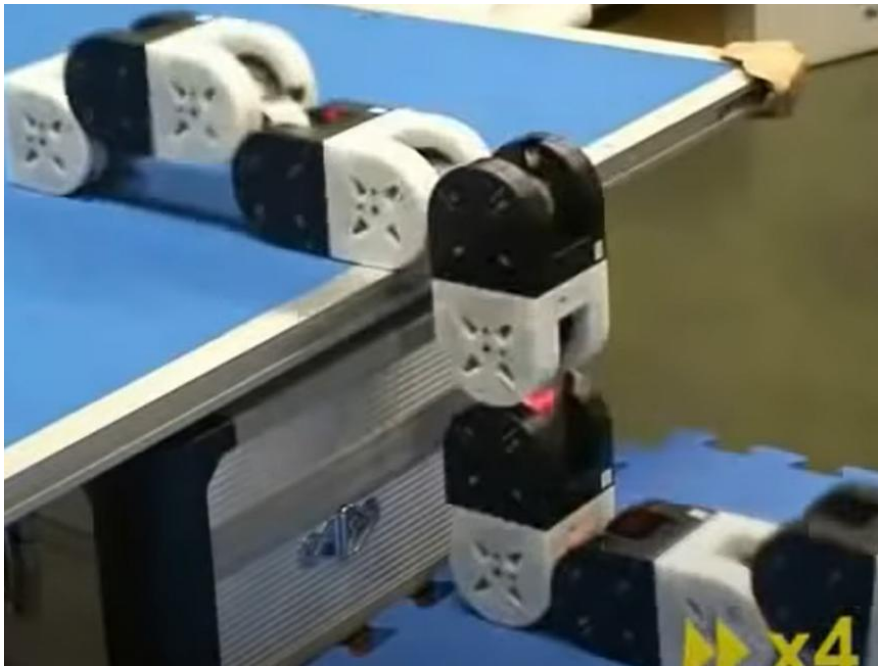
Locomotion animations (v) - caterpillar

http://www.youtube.com/watch?v=vfw4nduVU_E

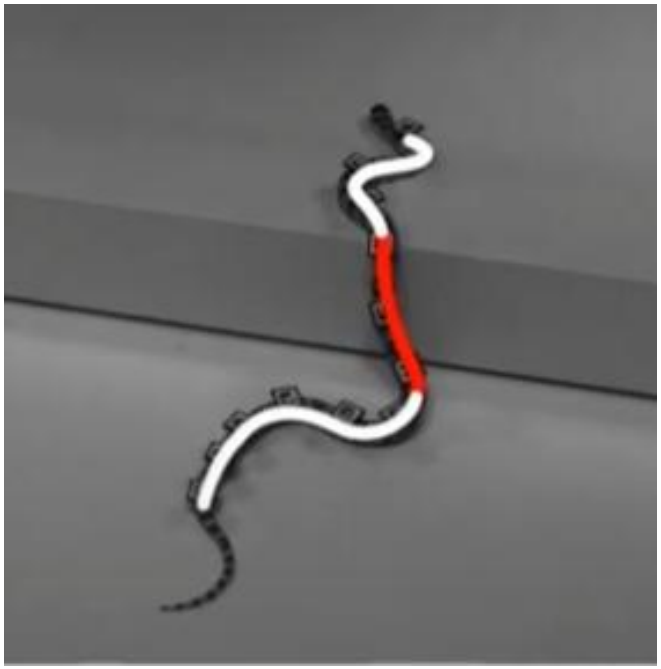
Locomotion animations (vi) - rolling

http://www.youtube.com/watch?v=f3rr1lcFe3Q&feature=related

Snake Morphing Adapt Locomotion…

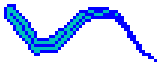


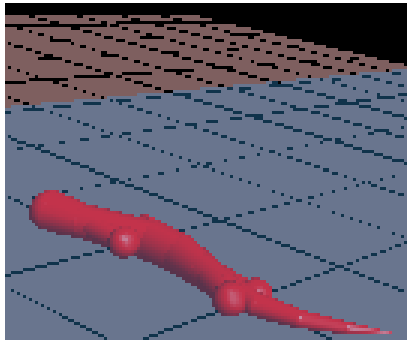
http://youtu.be/v6W-sEpJEqY



https://youtu.be/pv_MknD6jks

Locomotion animations – Biology inspired





Less common robots

Mtran Modular robot:
http://www.youtube.com/watch?v=4oSavAHf0dg
http://unit.aist.go.jp/is/frrg/dsysd/mtran3/mtran3.htm

Robotic Chair (reassembles itself):
http://youtu.be/t5pvZoZwzh0

LS3 - Legged Squad Support System ("dog"):
http://youtu.be/R7ezXBEBE6U
http://youtu.be/cNZPRsrwumQ

Ibot Robotic Wheel chair (climbs stairs):
http://youtu.be/O7otewMk9pc
https://youtu.be/9qYz2wsVKYE (more recent)

Origami Robot:
https://spectrum.ieee.org/origami-robot-folds-itself-up-does-cool-stuff-dissolves-into-nothing
https://youtu.be/f0CluQiwLRg
https://youtu.be/Awufipq9JnQ

Summary

- Many locomotion types…
  - According to environment…
- Most common and relevant to this course: Differential drive
  - OmniWheels a also easy!

~ Transmission ~

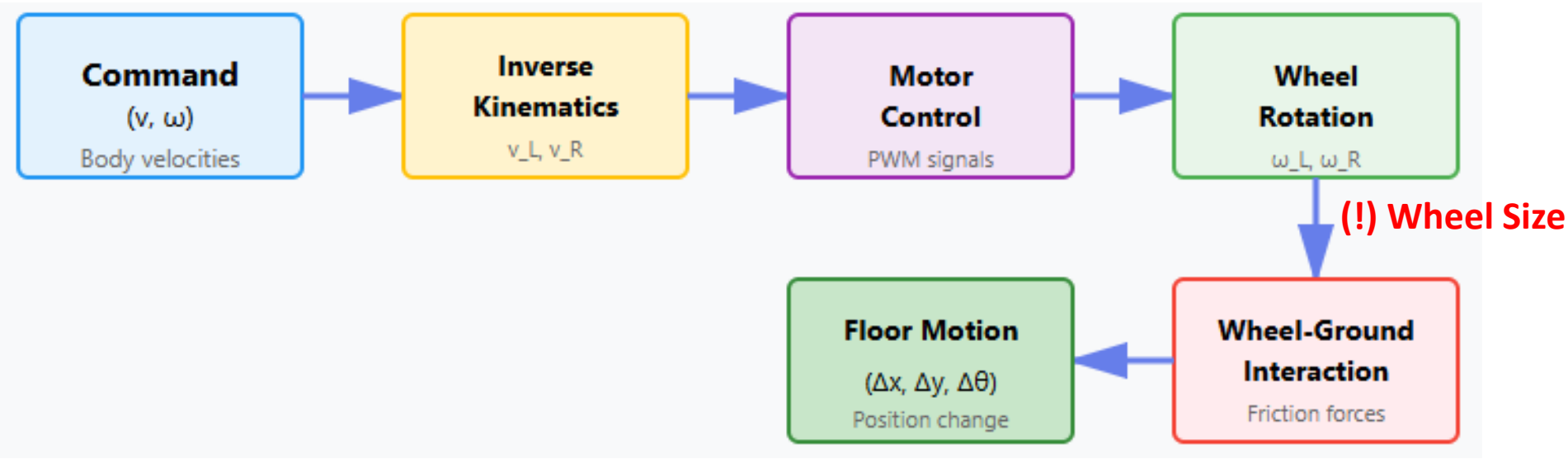
From (v, ω) to Floor Movement

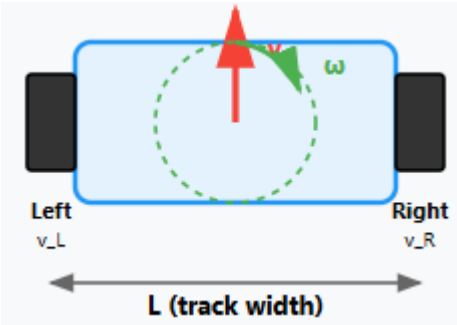
~ “Transmission” ~

From (v, ω) to Floor Movement

- Understand the transformation from body velocities to wheel velocities
- Apply inverse kinematics for differential drive robots
- Compute resulting position changes on the floor

The Complete Kinematic Chain





Inverse Kinematics Equations:

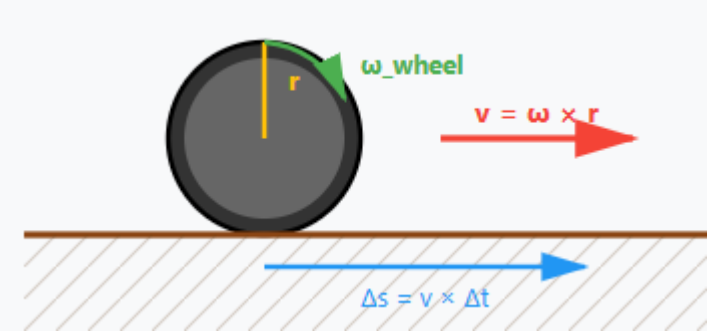
```
v_L = v - (ω × L/2)
v_R = v + (ω × L/2)
```

- **v**: Linear velocity at robot center
- **ω**: Angular velocity (rotation rate)
- **L**: Distance between left and right wheels
- Positive ω causes right wheel to move faster (left turn)

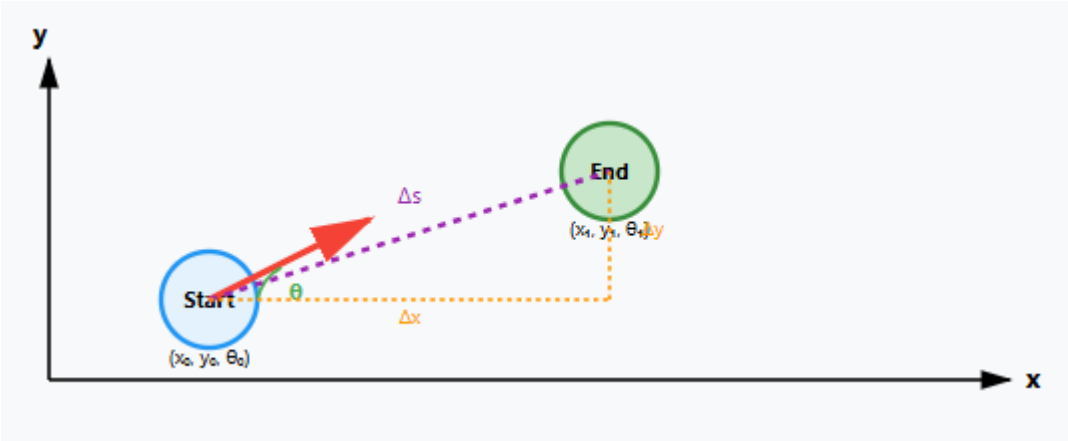
Wheel Rotation to Linear Motion:

$v_wheel = \omega_wheel \times r$

Distance traveled: $\Delta s = v \times \Delta t$



- Motor rotates wheel at angular velocity ω_wheel
- Wheel radius r converts rotation to linear velocity
- Friction between wheel and ground creates traction
- Robot moves forward with velocity $v = \omega \times r$



Position Update Equations (for time step Δt):

$$\Delta x = v \times \cos(\theta) \times \Delta t$$
$$\Delta y = v \times \sin(\theta) \times \Delta t$$
$$\Delta\theta = \omega \times \Delta t$$

New position:

$$x_1 = x_0 + \Delta x$$
$$y_1 = y_0 + \Delta y$$
$$\theta_1 = \theta_0 + \Delta\theta$$

Summary: Complete Flow (simplified)

1) Body to Wheels (Inverse Kinematics)

```
v_L = v - (ω × L/2)
v_R = v + (ω × L/2)
```

2) Wheels to Linear Motion

```
v_wheel = ω_wheel × r
(where ω_wheel = v_wheel / r from motor control)
```

3) Velocity Integration to Position

```
Δx = v × cos(θ) × Δt
Δy = v × sin(θ) × Δt
Δθ = ω × Δt
```

Summary: Complete Flow (simplified)

- This model assumes no wheel slip
- Non-holonomic constraint: robot cannot move sideways
- Forward kinematics reverses this process for odometry
- Real systems need calibration and slip compensation

$\Rightarrow$ Not counting electrical and mechanical poles associated with dynamics
$\Rightarrow$ Not caring about wheel imperfection
$\Rightarrow$ The two wheels are of slightly different size (perimeter)
$\Rightarrow$ The two {motor + gear} are different
$\Rightarrow$ Discuss: wheel rotation vs. ground speed (e.g. camera to the floor)

More information:

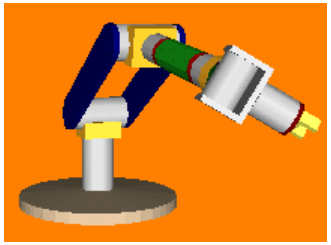
https://youtu.be/YxuJFQZh94Q

Kinematics & Dynamics

Degrees of Freedom

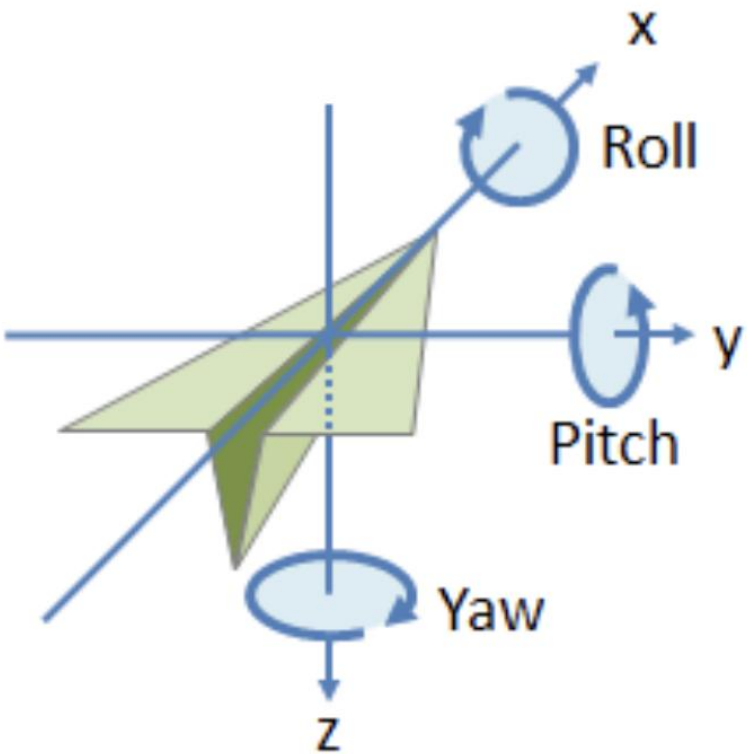
Mechanical engineering, aeronautical engineering, and robotics, the six DOFs of a rigid body have special names: This is a typical robot arm which has 7 DOF (including surge at the end of the arm).

1. Moving up and down (heaving);
2. Moving left and right (swaying);
3. Moving forward and backward (surging);
4. Tilting up and down (pitching);
5. Turning left and right (yawing);
6. Tilting side to side (rolling).



Obs: Euler angles!!!

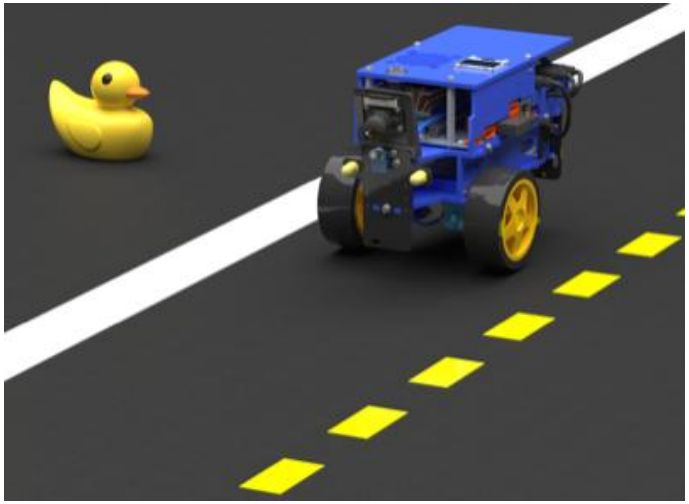
Degrees of Freedom (ii)



Axis of Rotation	Euler Angle Name	Euler Angle Symbol
x	Roll	u
y	Pitch	v
z	Yaw	w

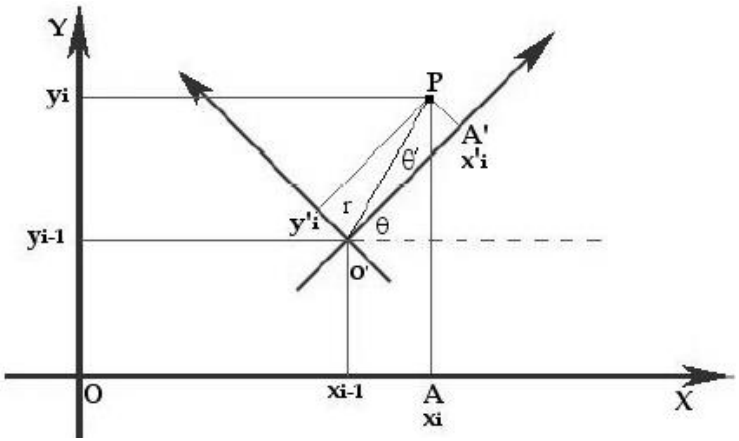
Differential Drive Floor Robot

("Wheelchair" drive)



https://duckietown.com/

2D movement - Dynamics



$$x_i = \overline{OA} = x_{i-1} + r*\cos(\theta+\theta') = x_{i-1} + r*\cos(\theta)*\cos(\theta) - r*\sin(\theta)*\sin(\theta) \quad (1)$$

$$y_i = \overline{AP} = y_{i-1} + r*\sin(\theta+\theta') = y_{i-1} + r*\sin(\theta)*\cos(\theta') + r*\cos(\theta)*\sin(\theta') \quad (2)$$

$$x_i' = \overline{O'A'} = r*\cos(\theta') . \quad (3)$$

$$y_i' = \overline{A'P} = r*\sin(\theta') . \quad (4)$$

$$x_i = x_i'\cos(\theta) - y_i'\sin(\theta) + x_{i-1} . \quad (5)$$

$$y_i = x_i'\sin(\theta) + y_i'\cos(\theta) + y_{i-1} . \quad (6)$$

$$\theta_i = \theta_I' + \theta_{i-1} . \quad (7)$$

$$x_i = d_i*\cos(\theta_i) + x_{i-1} . \quad (8)$$

$$y_i = d_i*\sin(\theta_i) + y_{i-1} . \quad (9)$$

Simulation of Mobile Robots in Virtual Environments
Jesús Savage, Emmanuel Hernández, Gabriel Vázquez, Humberto Espinosa, Edna Márquez

Position: $x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 .$

Speed: $\dfrac{dx(t)}{dt} = a_1 + 2a_2 t + 3a_3 t^2 .$

Acceleration: $\dfrac{dx(t)^2}{dt^2} = 2a_2 + 6a_3 t .$

2D Robot dynamic model

- State Vector:

$$X_r^T = [x_r\, y_r\, \theta_r\, v_{rt}\, v_{rn}\, \omega_r]$$

- Dinamics:

$$\dot{x}_r = v_{rt}\cos\theta_r - v_{rn}\sin\theta_r$$

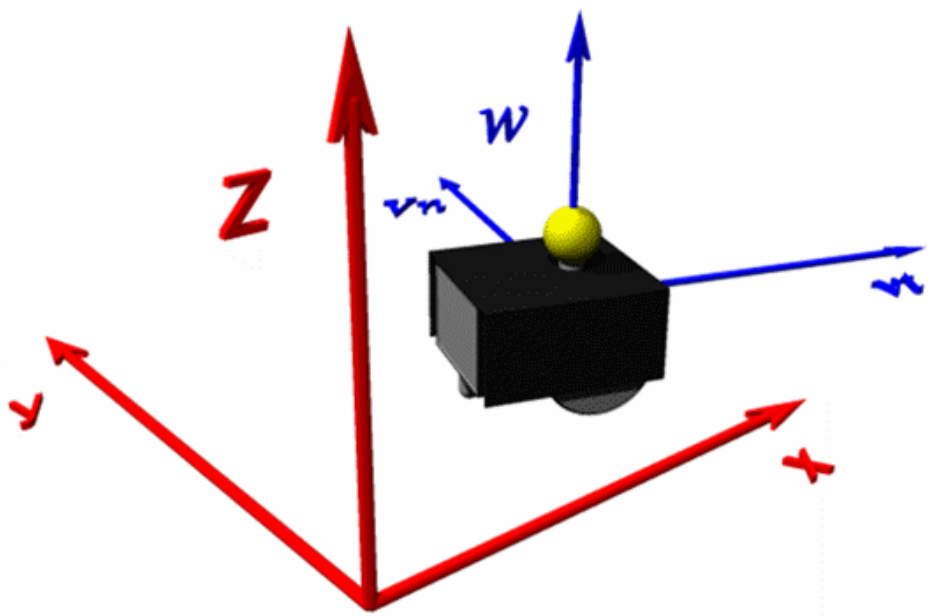
$$\dot{y}_r = v_{rt}\sin\theta_r + v_{rn}\cos\theta_r$$

$$\dot{\theta}_r = \omega_r$$

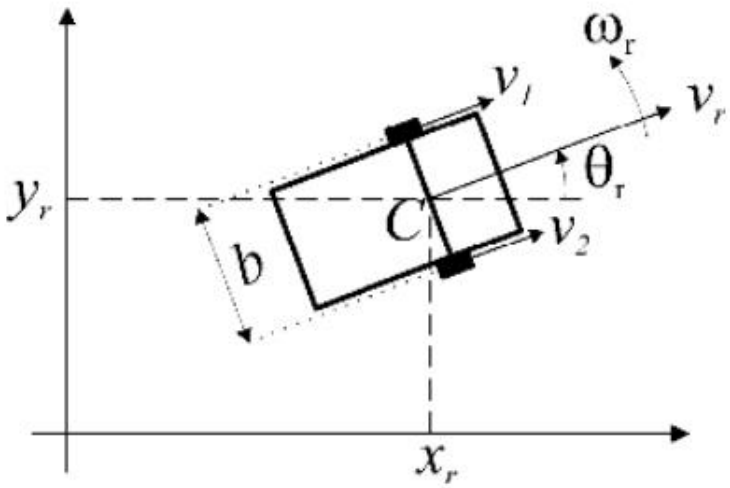
$$\dot{v}_{rt} = \alpha\left(v_{ref} - v_{rt}\right)$$

$$\dot{v}_{rn} = 0$$

$$\dot{\omega}_r = \gamma\left(\omega_{ref} - \omega_r\right)$$



$$\begin{cases} v = \frac{v_1+v_2}{2} \\ \omega = \frac{v_1-v_2}{b} \end{cases}$$



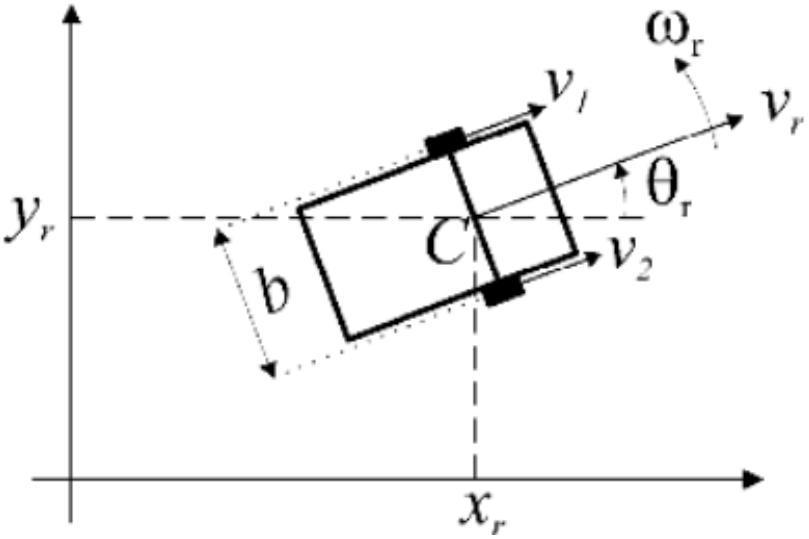
$$\begin{cases} v_2 = v_1 - \omega b \end{cases}$$

$$\begin{cases} 2v = v_1 + v_1 - \omega b \end{cases}$$

$$\begin{cases} v_1 = v + \omega b/2 \end{cases}$$

$$\begin{cases} v_1 = v + \omega b/2 \\ v_2 = v - \omega b/2 \end{cases}$$

Differential ground robot



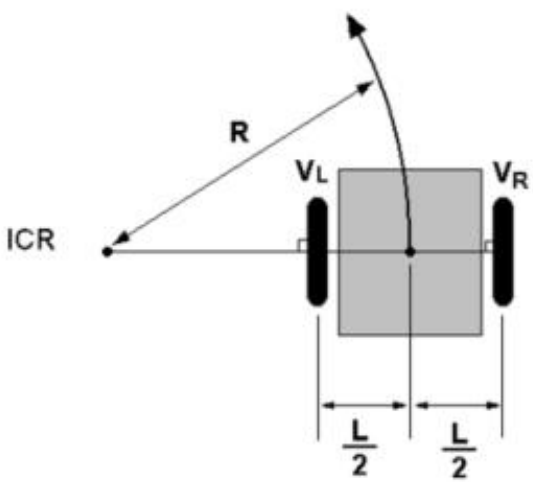
$$v_t = v_r$$

$$\begin{cases} v_t = \dfrac{(v_1 + v_2)}{2} \\ \omega_r = \dfrac{(v_1 - v_2)}{b} \end{cases}$$

"Pose" $\Rightarrow (x, y, \theta)$

$$\begin{cases} x_r(t+1) = x_r(t) + v_r(t) \cdot \cos\theta_r(t) \cdot \Delta t \\ y_r(t+1) = y_r(t) + v_r(t) \cdot \sin\theta_r(t) \cdot \Delta t \\ \theta_r(t+1) = \theta_r(t) + \omega_r(t) \cdot \Delta t \end{cases}$$

V, ω, v_L, v_R



$$(V_R - V_L)/\ L = V_R\ /(R + \frac{L}{2})$$

$$R = \frac{L}{2}\ \frac{V_R + V_L}{V_R - V_L}$$

R : Radius of rotation

- Straight motion

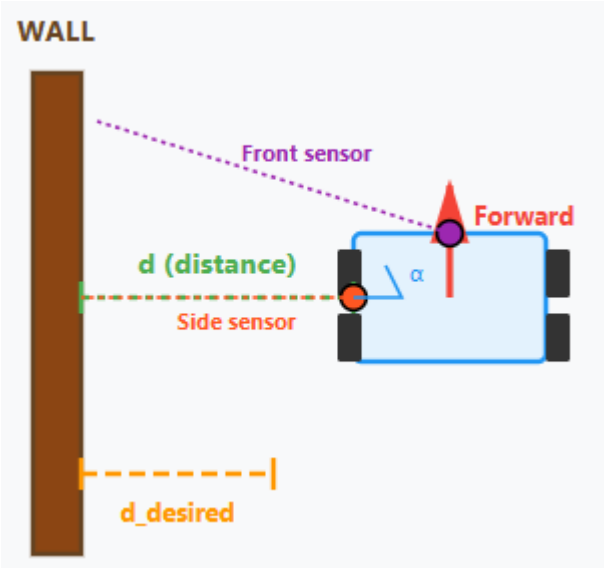
  R = Infinity → $V_R = V_L$

- Rotational motion

  R = 0 → $V_R = -V_L$

Wall Following

Wall Following



Key Measurements:

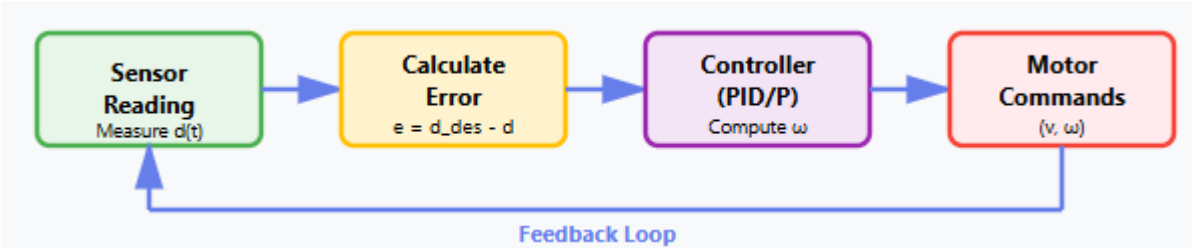
- **d**: Current perpendicular distance to wall (from side sensor)
- **d_desired**: Target distance to maintain from wall
- **α**: Robot's orientation angle relative to wall
- **Front sensor**: Detects obstacles ahead (for collision avoidance)
- **Side sensor**: Measures lateral distance to wall

`Error Signal:`

```
e(t) = d_desired - d(t)

e(t) > 0 → Robot too far from wall
(turn left)
e(t) < 0 → Robot too close to wall
(turn right)
e(t) ≈ 0 → Robot at desired distance
(go straight)
```

Wall Following



Control Laws:

1. Proportional (P) Controller (Simple):

$\omega(t) = K_p \times e(t)$

$v(t) = v_{constant}$

where K_p is the proportional gain

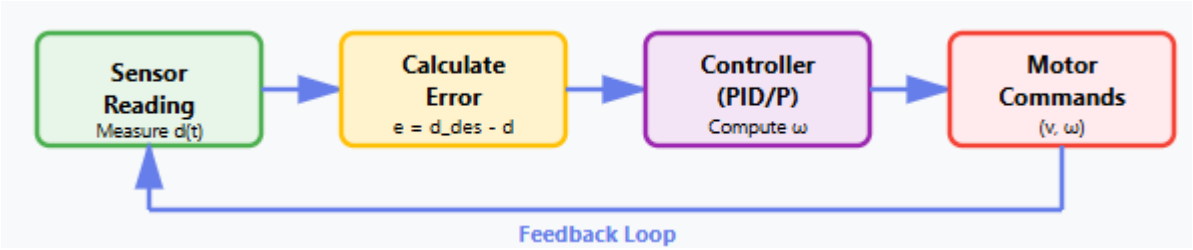
2. PD Controller (Better Stability):

$\omega(t) = K_p \times e(t) + K_d \times \dot{e}(t)$

$v(t) = v_{base} - K_v \times |e(t)|$

where K_d is the derivative gain (damping)

Wall Following



Behavior Summary:

If robot is too far from wall (e > 0):

•Positive angular velocity ω → Turn left (toward wall)

•Reduce forward velocity v slightly

If robot is too close to wall (e < 0):

•Negative angular velocity ω → Turn right (away from wall)

•Reduce forward velocity v slightly

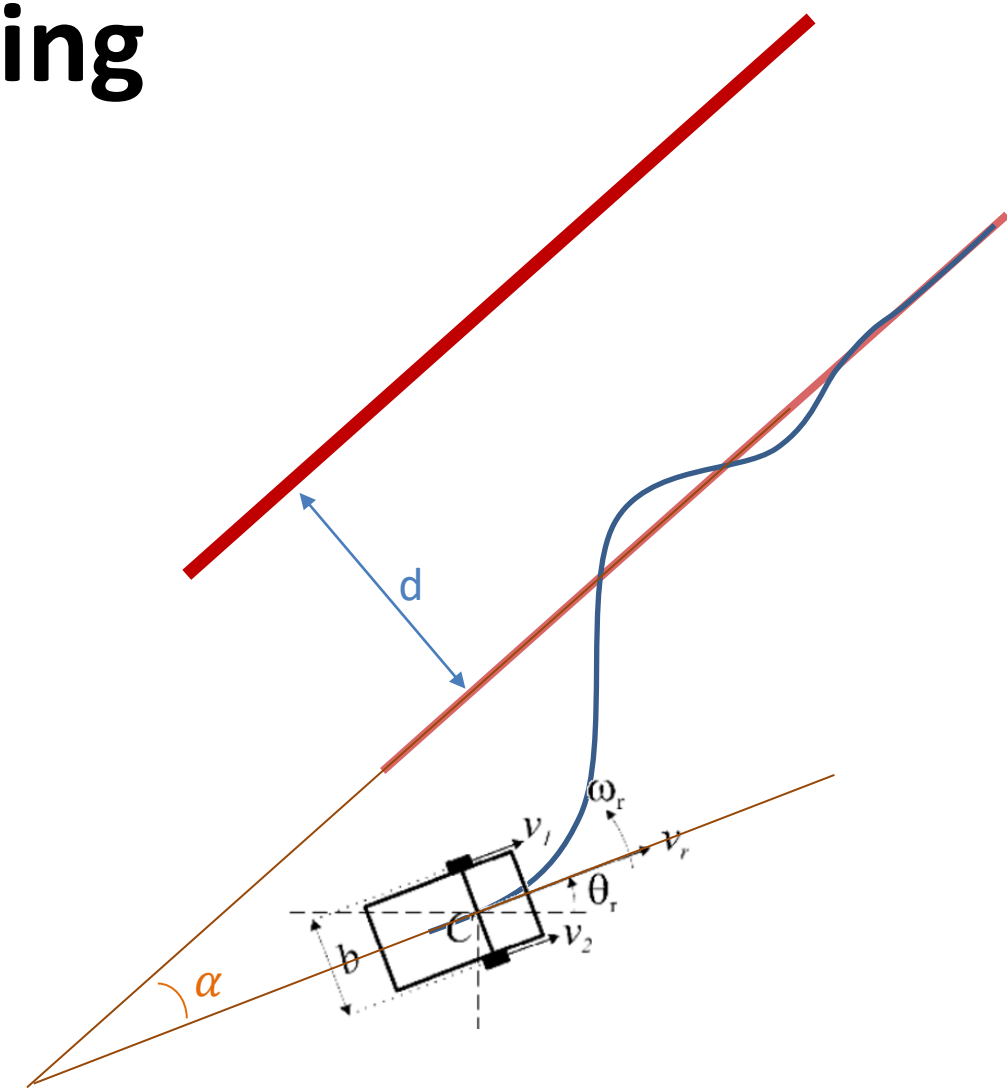
If robot is at correct distance (e ≈ 0):

•Zero angular velocity ω → Go straight

•Maintain nominal forward velocity v

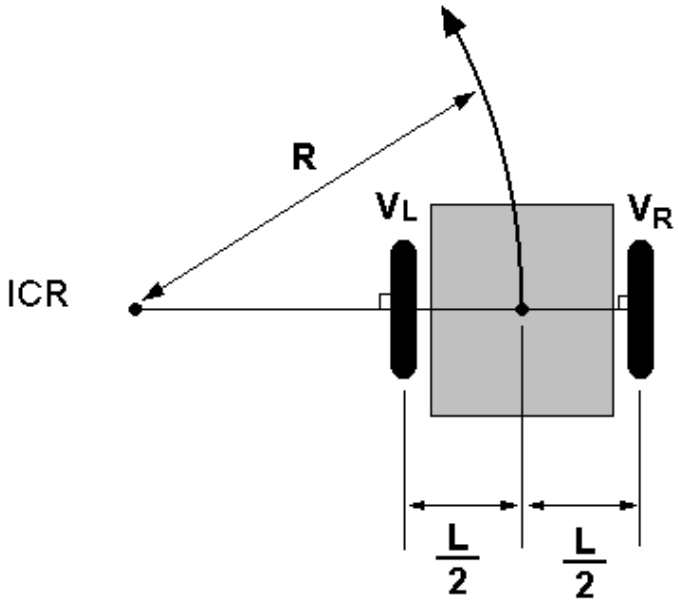
Wall following

- **Always use floating point and SI units**
- **Many solutions…**
- **Example of a simple control:**
  - Have robot rotate proportionally to the difference of angle (robot and Wall to follow)
  - $\omega_r(t) = k \cdot \alpha(t-1)$ (try k and –k)



More information

Curv. Radius



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$$R = \frac{L}{2}\ \frac{V_R + V_L}{V_R - V_L}$$

R : Radius of rotation

- Straight motion

  R = Infinity $\qquad V_R = V_L$

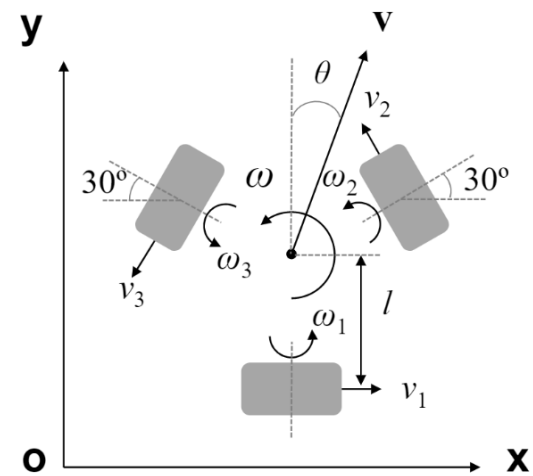
- Rotational motion

  R = 0 $\qquad V_R = -V_L$

YARP

- https://github.com/asousa-feup/YARP

Omni



Velocity in the body coordinates :

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} v\sin\theta \\ v\cos\theta \end{bmatrix}$$

Velocity of the wheels :

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} r\omega_1 \\ r\omega_2 \\ r\omega_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & l \\ -\sin 30^\circ & \cos 30^\circ & l \\ -\sin 30^\circ & -\cos 30^\circ & l \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix}$$

r : **Radius of the wheel**

Relationship between the world and body coordinates :

$$\underbrace{\begin{bmatrix} v_x \\ v_y \end{bmatrix}}_{\text{Velocity in the body coordinates}} = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix} \underbrace{\begin{bmatrix} v_x^w \\ v_y^w \end{bmatrix}}_{\text{Velocity in the world coordinates}}$$

ϕ : **Heading direction of the robot**

Velocity of the wheels :

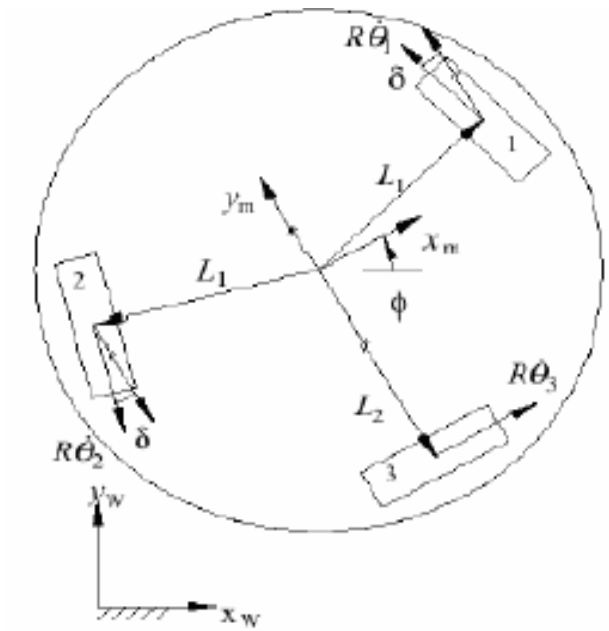
$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & l \\ -\sin 30^\circ & \cos 30^\circ & l \\ -\sin 30^\circ & -\cos 30^\circ & l \end{bmatrix} \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_x^w \\ v_y^w \\ \omega \end{bmatrix}$$

$$= \begin{bmatrix} \cos\phi & \cos\phi & l \\ -\sin(\phi+30^\circ) & \cos(\phi+30^\circ) & l \\ \sin(\phi-30^\circ) & -\cos(\phi-30^\circ) & l \end{bmatrix} \begin{bmatrix} v_x^w \\ v_y^w \\ \omega \end{bmatrix}$$

where

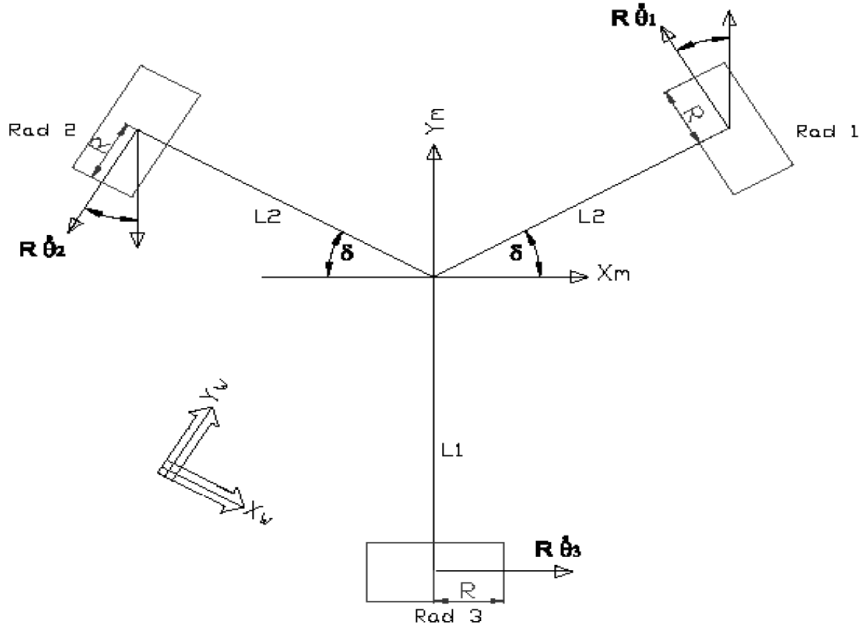
$$\phi = \int \omega \, dt$$

Omniwheels



$$\begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{Bmatrix} = \frac{1}{R} \begin{bmatrix} -\sin(\delta) & -\cos(\delta) & L_1 \\ -\sin(\delta) & \cos(\delta) & L_1 \\ 1 & 0 & L_2 \end{bmatrix} \begin{Bmatrix} \dot{x}_m \\ \dot{y}_m \\ \dot{\phi} \end{Bmatrix}$$

Omniwheels (ii)



$$\begin{Bmatrix} \dot{\Theta}_1 \\ \dot{\Theta}_2 \\ \dot{\Theta}_3 \end{Bmatrix} = \frac{1}{R} \begin{bmatrix} -\sin(\delta+\Phi) & \cos(\delta+\Phi) & L_1 \\ -\sin(\delta-\Phi) & -\cos(\delta-\Phi) & L_1 \\ \cos(\Phi) & \sin(\Phi) & L_2 \end{bmatrix} \begin{Bmatrix} \dot{x}_w \\ \dot{y}_w \\ \dot{\Phi} \end{Bmatrix}$$

Robotic Locomotion & Kinematics & Wall Following

Armando Sousa

Jorge Ribeiro